



Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

A-level MATHEMATICS

Paper 3

Thursday 19 June 2025

Afternoon

Time allowed: 2 hours

Materials

- You must have the AQA Formulae for A-level Mathematics booklet.
- You should have a graphical or scientific calculator that meets the requirements of the specification.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer each question in the space provided for that question.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do not write outside the box around each page or on blank pages.
- Show all necessary working; otherwise marks for method may be lost.
- Do all rough work in this book. Cross through any work that you do not want to be marked

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 100.

Advice

- Unless stated otherwise, you may quote formulae, without proof, from the booklet.
- You do not necessarily need to use all the space provided.

For Examiner's Use

Question	Mark
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TOTAL	



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Section A

Answer **all** questions in the spaces provided.

- 1** A student states that the product of two irrational numbers is always irrational.

One of the options is a counter example that shows the student's statement is incorrect.

Identify the counter example.

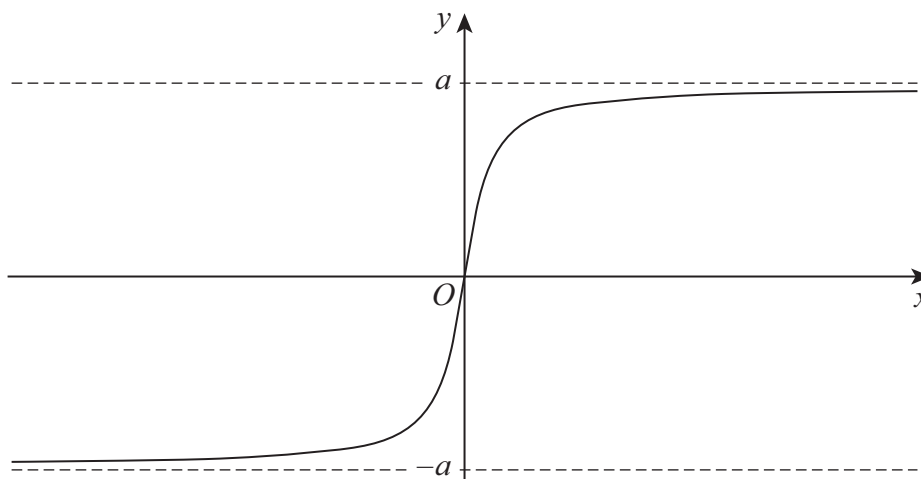
Circle your answer.

[1 mark]

$$e \times 0 = 0 \qquad \frac{1}{\pi} \times \pi = 1 \qquad \frac{1}{\sqrt{2}} \times \sqrt{6} = \sqrt{3} \qquad 2 \times \sqrt{3} = \sqrt{6}$$



- 2 The diagram shows the graph of $y = \arctan x$



The graph has horizontal asymptotes at $y = -a$ and $y = a$

State the value of a

Circle your answer.

[1 mark]

$$\frac{\pi}{4}$$

$$1$$

$$\frac{\pi}{2}$$

$$\pi$$

Turn over for the next question

Turn over ►



3 The function f is defined by

$$f(x) = e^x \text{ for } x \in \mathbb{R}$$

Identify which one of the following statements describes the function f

Tick (✓) **one** box.

[1 mark]

Decreasing and concave

☐

Decreasing and convex

☐

Increasing and concave

☐

Increasing and convex

☐

4 (a) State the value of

$$\log_a a^2$$

where $a > 0$

[1 mark]

4 (b) Find the value of

$$\log_a \sqrt{a} - \log_a \frac{1}{a^2}$$

where $a > 0$

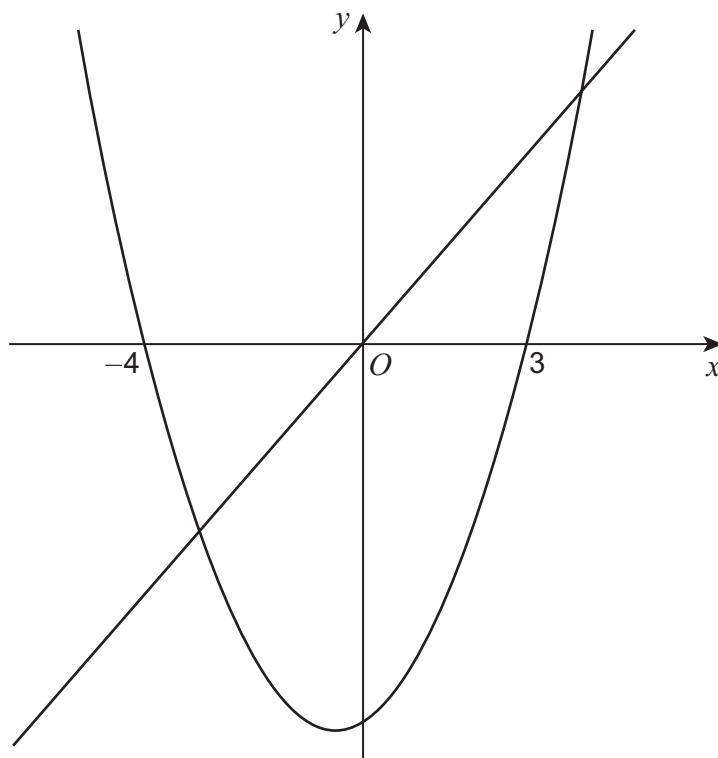
[2 marks]

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- 5 The diagram shows the graphs of $y = 2x$ and $y = (x - 3)(x + 4)$.



The region R is defined by the inequalities

$$(x - 3)(x + 4) \leq y \leq 2x \text{ and } x \leq 0$$

Shade the region R on the diagram.

[2 marks]



- 6** The first four terms in ascending powers of x of the binomial expansion of

$$(2 - 3x)^5$$

are given by

$$32 + px + qx^2 - 1080x^3$$

where p and q are constants.

- 6 (a)** Find the value of p and the value of q

[2 marks]

- 6 (b)** Hence find an approximation for 1.94^5

Give your answer to five decimal places.

[2 marks]

Turn over ►



7 Solve the equation

$$\sum_{r=1}^3 (ar + 5) = 57$$

to find the value of a

[3 marks]



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0 9

8 (a) The expression

$$\frac{x}{2x^2 + 3x + 1}$$

can be written in the form

$$\frac{A}{x+1} + \frac{B}{2x+1}$$

Find the value of A and the value of B

[3 marks]



8 (b) Use your answer to part (a) to show that

$$\int_0^4 \frac{x}{2x^2 + 3x + 1} dx = \ln q$$

where q is a rational number to be found.

[4 marks]

Turn over ►



- 9 (a)** Describe a sequence of **two** transformations which maps the graph with equation

$$y = \frac{1}{x}$$

onto the graph with equation

$$y = \frac{3}{x-4}$$

[2 marks]

- 9 (b)** State the equation of the vertical asymptote of the graph with equation

$$y = \frac{3}{x-4}$$

[1 mark]



- 9 (c)** A student is attempting to use a change of sign to determine if the equation

$$\frac{3}{x-4} = x$$

has a solution between 3 and 5

The student **correctly** writes

$$\frac{3}{x-4} = x \Leftrightarrow \frac{3}{x-4} - x = 0$$

$$\text{Let } f(x) = \frac{3}{x-4} - x$$

$$f(3) = -6 < 0 \text{ and } f(5) = -2 < 0$$

The student then **incorrectly** states:

“Since there is no change of sign, there is no solution between 3 and 5”

Give two reasons why the student’s argument is invalid.

[2 marks]

Reason 1: _____

Reason 2: _____

Turn over for the next question

Turn over ►



$$2 \sin x + 5 \cos x$$
$$R \sin (x + \alpha)$$

[3 marks]

[illegible]
$$T = 14.2 - (2 \sin d^\circ + 5 \cos d^\circ)$$

- T is the temperature of the water in $^{\circ}\text{C}$
- d is the number of days after 1 January 2010



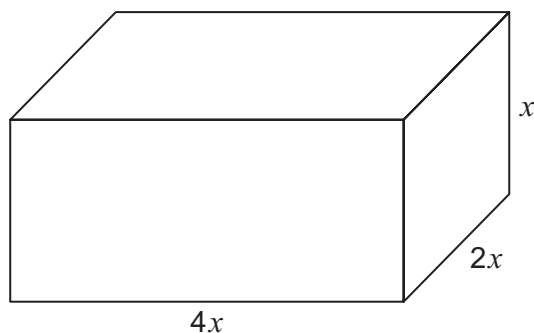
Give your answer to one decimal place.

Give your answer to the nearest whole number.

[illegible]

A block of ice is melting.

At time t minutes, the block is in the shape of a cuboid with dimensions of $4x$, $2x$ and x , as shown in the diagram.



All measurements are in centimetres.

11 (a) The volume, $V \text{ cm}^3$, of the block of ice **decreases** at a rate which is proportional to its surface area.

When $x = 4$ the volume of the block of ice is decreasing at a rate of 7 cm^3 per minute.

Show that

$$\frac{dV}{dt} = -0.4375x^2$$

[4 marks]

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11 (b) Find $\frac{dV}{dx}$ in terms of x

[2 marks]

11 (c) (i) Using the results from parts **(a)** and **(b)**, find $\frac{dx}{dt}$

[2 marks]

11 (c) (ii) Interpret, in context, your answer to part **(c)(i)**.

[2 marks]

Turn over ►



Given that the real numbers a , b , and c are such that

Use proof by contradiction to show that if c is irrational then at least one of a or b is irrational.

[illegible]

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Section B

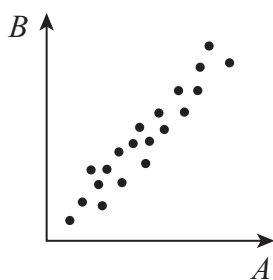
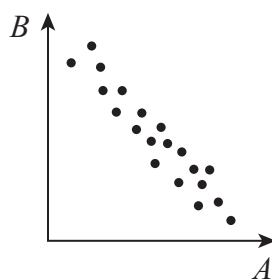
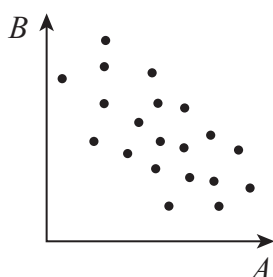
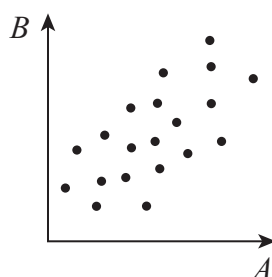
Answer **all** questions in the spaces provided.

13

The variables A and B have a product moment correlation coefficient of 0.47

Identify which one of the scatter diagrams shows the relationship between A and B .

Tick (✓) **one** box.

[1 mark]☐☐☐☐

14

It is given that $X \sim N(9, 1.5^2)$ and $P(a \leq X \leq b) \approx 95\%$

Identify the correct pair of values of a and b .

Tick (✓) **one** box.

[1 mark]

$a = 4.5$ and $b = 13.5$

☐

$a = 6$ and $b = 12$

☐

$a = 7$ and $b = 11$

☐

$a = 7.5$ and $b = 10.5$

☐

Turn over for the next question

Turn over ►



15 The proportion of diners at a restaurant who are vegan is 20%

The number of diners at the restaurant who are vegan can be modelled by a binomial distribution.

15 (a) In a random sample of 25 diners:

15 (a) (i) find the probability that no diner is a vegan.

[1 mark]

15 (a) (ii) find the probability that fewer than 6 diners are vegan.

[1 mark]

15 (a) (iii) find the probability that at least 9 diners are vegan.

[2 marks]

15 (b) State one assumption necessary for the binomial distribution to be valid in the context of this question.

[1 mark]



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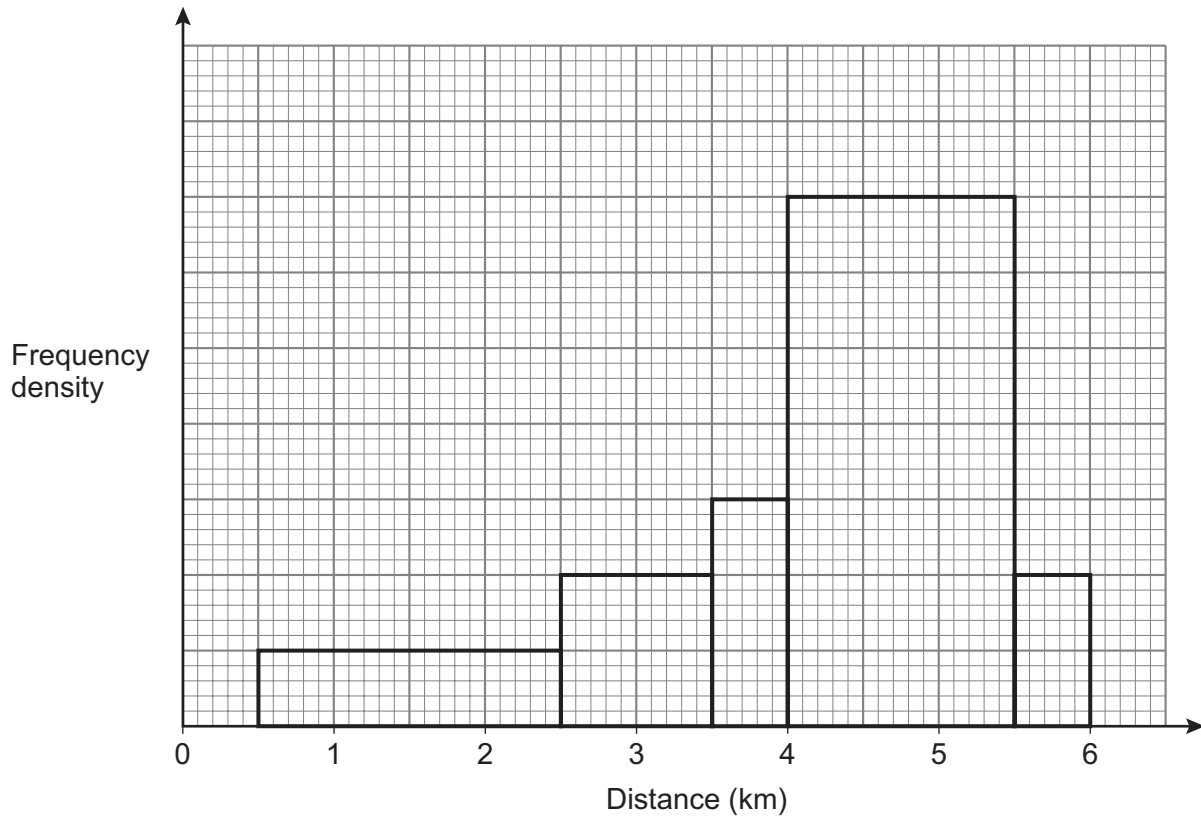
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16

The distance, in kilometres, travelled from home to work by 340 individuals is shown in the histogram.



- 16 (a) (i)** Estimate the number of individuals who travel between 3 km and 5 km from home to work.

Fully justify your answer.

[4 marks]



16 (a) (ii) Give one reason why your answer to part **(a)(i)** is only an estimate.

[1 mark]

16 (b) The histogram shows that the data is skewed.

State the type of skewness.

[1 mark]

16 (c) Explain why frequency density is used in this histogram.

[1 mark]

Turn over ►



17 A maths teacher holds a revision session every Wednesday.

In a random sample of eight Wednesdays, the number of students who attended is listed below.

2 4 3 1 4 5 2 3

17 (a) Find the mean of these eight values.

[1 mark]

17 (b) Find the variance of these eight values.

[1 mark]

17 (c) The teacher believes that the number of students who attended a revision session each Wednesday can be modelled by the binomial distribution $B(30, 0.1)$.

Comment on whether the mean and variance found in parts **(a)** and **(b)** support the teacher's belief.

Fully justify your answer.

[3 marks]



17 (d) The teacher wanted to decide if a Saturday morning revision session was worth doing.

The teacher asked the first 10 students who entered the classroom if they would attend a Saturday session.

17 (d) (i) Name this method of sampling.

[1 mark]

17 (d) (ii) Describe **one advantage** of this method of sampling.

[1 mark]

Turn over for the next question

Turn over ►



18 The cholesterol level, X , of an adult can be modelled by a normal distribution with mean 5.7 mmol/l and standard deviation 1.2 mmol/l.

18 (a) The lower quartile of X is a and the upper quartile is b

It is given that the $P(X \leq a) = 0.25$ and $P(X \geq b) = 0.25$

Find the interquartile range of X

[4 marks]



A group of scientists believe that a new dietary supplement can reduce the mean cholesterol level.

After the trial, their cholesterol level was found to have a mean of 5.6 mmol/l.

It can be assumed that the cholesterol level after taking the dietary supplement follows a normal distribution with an unchanged variance.

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- 19** Three events X , Y and Z are such that

$$P(X) = P(Y) = 0.38$$

$$P(X \cup Z) = 0.5$$

$$P[(X \cup Y)'] = 0.35$$

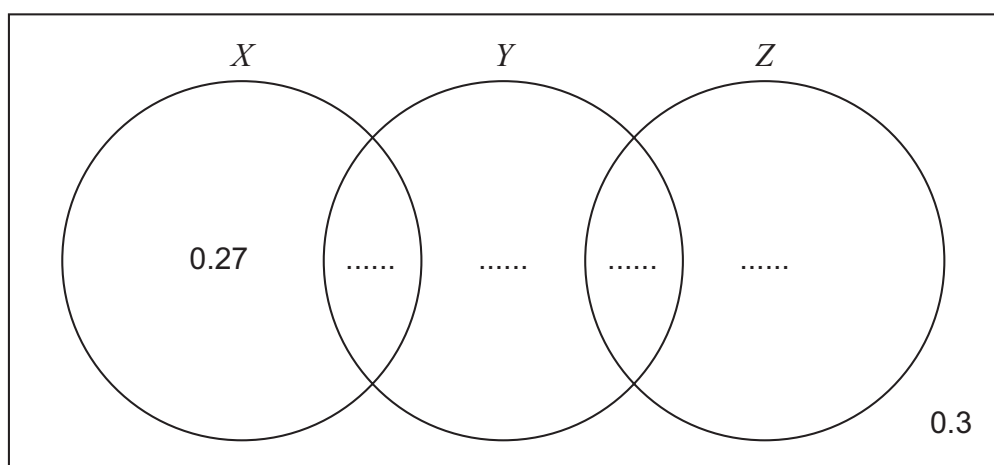
X and Z are mutually exclusive.

- 19 (a)** Find $P(Z)$

[1 mark]

- 19 (b)** Complete the Venn diagram.

[3 marks]



19 (c) Find $P[(X \cap Y) \cup (Y \cap Z)]$

[2 marks]

19 (d) Find $P(Y' | Z')$

[2 marks]

19 (e) Show that the events Y and Z are **not** independent of each other.

[2 marks]

Turn over ►



20 The oxides of nitrogen emissions, F g/km, of a car registered in 2000 can be modelled by a normal distribution with mean 0.41 and standard deviation 0.07

20 (a) Find $P(F < 0.39)$

[1 mark]

20 (b) Find $P(0.3 < F < 0.5)$

[1 mark]

20 (c) (i) Find $P(F > 0.6)$

[1 mark]

20 (c) (ii) Explain why $P(F \geq 0.6) = P(F > 0.6)$ in this model.

[2 marks]



- 20 (d)** The oxides of nitrogen emissions, F g/km, of a car in 2010 can be modelled by a normal distribution with mean 0.36 and standard deviation 0.09

Compare the oxides of nitrogen emissions in 2000 with those in 2010.

[2 marks]

- 20 (e)** A researcher collected data on the oxides of nitrogen emissions from the Large Data Set for cars registered in 2002 and 2016.

The researcher cleaned the data by removing the details for some cars.

Using your knowledge of the Large Data Set, give one reason why the researcher cleaned the data.

[1 mark]

- 20 (f)** The researcher wished to make a comparison between the oxides of nitrogen emissions for a sample of Nissan cars in their local town from 2016 with the data from all cars registered in the same year in the Large Data Set.

Using your knowledge of the Large Data Set, give one reason why a meaningful comparison could **not** be made.

[1 mark]

END OF QUESTIONS



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[illegible]

